# MIN-MARGINAL POWER INDICES FOR MULTI-OBJECTIVE FEATURE SELECTION

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## ABSTRACT

Game-theoretic feature selection usually aggregates a feature's marginal contributions across coalitions by a mean, which can conceal a context in which adding that feature reverses an otherwise acceptable decision. Whether a feature should be removed depends on whether it is harmful in such a context: filtering harmful features can improve single-task models, whereas blind deletion tends to degrade multi-task models in which features can harm one another. Building on minimum marginal aggregation, we specialize it to a binary utility-threshold voting game and call the resulting quota-indexed support statistic the Min-Marginal Power Index (MinPPI). For a fixed utility quota $\alpha$, exact MinPPI lies in $\{-1, 0, +1\}$, classifying each feature as harmful, neutral, or decisive at the worst-case level from witness-certified threshold crossings alone, without a second score cutoff; its minimum Shapley–Shubik and minimum Banzhaf analogues coincide at the support level. We pair exact computation with finite-budget estimators whose detection guarantees are stated only as conditional support-mass bounds, and pass MinPPI's harmful candidates through a validation gate before removal or multi-objective search. Across six benchmark datasets over five frozen seeds, MinPPI produced heterogeneous, low-powered evidence: it flagged harmful features on some datasets and none on others, and its validation-gated removals and prefiltering showed no significant average advantage over strong baselines in downstream search. MinPPI is therefore best understood as a conservative feature-deletion strategy: it removes only witness-certified harmful features under a validation gate, performs particularly well on datasets with abundant removable structure—removing up to $44\%$ of features at negligible utility cost and improving downstream search on Madelon—and correctly removes little or nothing where all features carry signal. We argue that worst-case support statistics are the right object for deciding when feature removal is evidence-backed.

## 1 INTRODUCTION

Feature selection balances predictive utility against the number of retained variables and is therefore naturally multi-objective (Guyon & Elisseeff, 2003; Chandrashekar & Sahin, 2014; Xue et al., 2016). Cooperative-game methods treat features as players and a fitted model's utility as the coalition value, a formulation introduced by Cohen et al. (2007). The Shapley value and Banzhaf index, popularized for model explanation by SHAP (Lundberg & Lee, 2017), summarize average marginal contributions (Shapley, 1953; Banzhaf, 1965), and related selectors include Banzhaf-FS (Karavas et al., 2017), Powershap (Verhaeghe et al., 2022), and ShapG (Zhao et al., 2025). Evolutionary methods such as NSGA-II take a different route, approximating Pareto fronts over the combinatorial feature space.

Removing a feature is beneficial only if the feature is harmful in some context: for single-task models, filtering harmful features can improve performance (Lippl & Lindsey, 2024), whereas in multi-task models features can harm one another and blind deletion often degrades the shared model (Siam et al., 2026). Both observations argue for certifying harmfulness from evidence rather than inferring it from an aggregate score. A mean marginal contribution answers whether a feature helps on average, not whether there exists a coalition in which the feature reverses an otherwise acceptable decision. MinShap replaces averaging by minimum aggregation for general conditional-importance functionals (Zheng & Raskutti, 2026), but its continuous score still needs a separate cutoff to de-

clare a feature harmful. We specialize the same aggregation to a binary utility-threshold game, which supplies trinary support-level semantics: an observed $-1$ is existence-certified by an evaluated threshold-crossing coalition and, for a configured quota $\alpha$, the exact output needs no additional real-valued score cutoff. The quota remains a substantive user choice and is not removed by the construction.

Our contributions are fourfold. First, we define the MinPPI specialization and establish its trinary range, quota-conditioned worst-case semantics, and support-level coincidence of its minimum Shapley–Shubik and minimum Banzhaf analogues. Second, we separate exact computation from finite-budget estimation: observed harmful labels are witness-certified, while missed witnesses remain possible, and the guidance bound is explicitly conditional on the perturbation distribution assigning positive mass to a harmful witness. Third, we use $\alpha$ to generate quota-indexed compression–utility operating points and pass candidate removals through a validation-only gate before test evaluation or downstream search. Fourth, we report a provenance-checked, five-seed analysis on six benchmark datasets, positioning MinPPI as a conservative feature-deletion strategy whose strongest results arise on datasets with abundant removable structure. Its evidence is deliberately mixed: removal methods are not statistically separated, prefiltering yields a positive but non-significant average hypervolume change, and guidance behavior depends on dataset, estimator, and utility.

## 2 METHOD

### 2.1 UTILITY-THRESHOLD VOTING GAME

Let $N = \{1, \ldots, n\}$ denote the analyzed features. For $S \subseteq N$, let $\mathcal{M}_S$ be the fixed model family fitted using $S$, and let $U(\mathcal{M}_S)$ be a configured real-valued, higher-is-better utility. The empty coalition uses a constant predictor derived only from the training response: class priors for classification and the training mean for regression. With full-feature utility $U_{\text{full}} = U(\mathcal{M}_N) > 0$ and quota $\alpha \in (0, 1]$, define

$$v_\alpha(S) = \mathbb{1}[U(\mathcal{M}_S) \geq \alpha U_{\text{full}}]\,. \tag{1}$$

Unlike a classical monotone simple game (Owen, 1972; Straffin, 1977), $v_\alpha$ need not be monotone: adding a feature can push utility just below the quota, and this single threshold flip is exactly the event MinPPI certifies. Its marginal is $\Delta_i(S) = v_\alpha(S \cup \{i\}) - v_\alpha(S) \in \{-1, 0, +1\}$.

### 2.2 MIN-MARGINAL POWER INDICES

For a permutation $\pi$ of $N$, let $S^i_\pi$ be the features preceding $i$. We define

$$\mathrm{MinSS}_i = \min_\pi \Delta_i(S^i_\pi), \tag{2}$$

$$\mathrm{MinBz}_i = \min_{S \subseteq N \setminus \{i\}} \Delta_i(S). \tag{3}$$

**Proposition 1 (exact semantics).** For every feature, $\mathrm{MinSS}_i = \mathrm{MinBz}_i \in \{-1, 0, +1\}$. The value is $-1$ exactly when some coalition is winning before $i$ is added and losing afterward; it is $+1$ exactly when every coalition is losing before $i$ is added and winning afterward; otherwise it is $0$. The value $-1$ is self-certifying: an evaluated coalition witnesses the flip from winning to losing, so the harmful label requires no additional score threshold. We call these classes *harmful*, *decisive*, and *neutral at the worst-case level*, respectively, always relative to the fixed model family, data split, utility, and quota. The last term means neither harmful nor universally decisive in the exact game; it does not mean that the feature has zero predictive value. The proof is in Appendix A.

Every coalition $S \subseteq N \setminus \{i\}$ occurs as $S^i_\pi$ for at least one permutation, so the two support-level minima have the same distinct arguments. This does not assert equality of the standard weighted Shapley–Shubik (Shapley & Shubik, 1954) and Banzhaf (Banzhaf, 1965) indices. We write $\mathrm{MinPPI}_i$ for either exact minimum analogue.

### 2.3 FINITE-BUDGET ESTIMATION

Exact evaluation requires at most $2^n$ distinct coalition utilities and $O(n2^{n-1})$ marginal comparisons. It is practical only at small $n$. For each feature, **Algorithm A** first samples a coalition size uniformly

from $\{0, \ldots, n-1\}$ and then samples a subset of that size uniformly without replacement from $N \setminus \{i\}$, in the permutation-sampling spirit of Shapley-effects estimation (Song et al., 2016). For $K \geq 1$, it returns the minimum observed marginal over the frozen schedule $\mathcal{S}_K$:

$$\widehat{\mathrm{MinPPI}}_i = \min_{S \in \mathcal{S}_K} \Delta_i(S) \geq \min_{S \subseteq N \setminus \{i\}} \Delta_i(S). \tag{4}$$

Thus an observed $-1$ certifies a harmful witness, whereas a sampled $0$ or $+1$ does not exclude an unseen smaller marginal.

**Algorithm B** uses training-only mutual information and absolute correlation—along the lines of ShapG's correlation-graph guidance (Zhao et al., 2025)—to construct a seed coalition: up to ten correlation neighbors are included and then padded from the mutual-information ranking toward $\min(20, \lfloor n/3 \rfloor)$. In each trial, every seed member is removed independently with probability .3; an addition count is drawn uniformly from $\{0, \ldots, \min(5, n-1)\}$ and that many currently absent features are taken without replacement from a random order. The resulting support is restricted by this construction. Conditional on the training data and frozen guidance but before drawing the random schedule, let $P_i$ denote the distribution of one perturbation. If a harmful witness $S_i^*$ has mass $\rho_i = P_i(S_i^*) > 0$ and the $K_{\mathrm{adv}} \geq 1$ draws are independent, then

$$\Pr[\text{detect a harmful witness}] \geq 1 - (1 - \rho_i)^{K_{\mathrm{adv}}}. \tag{5}$$

This geometric statement does not imply that MI-correlation guidance gives larger $\rho_i$ than random sampling. To make the support probabilities in Figure 2(a,b) auditable, consider $N = \{f_0, \ldots, f_4\}$ and the analytically specified game

$$v(S) = \mathbb{1}\big[f_0 \in S \ \wedge \ \big(f_3 \in S \ \vee \ f_4 \in S \ \vee \ (f_2 \in S \wedge f_1 \notin S)\big)\big]. \tag{6}$$

Feature $f_1$ has the harmful witness $S^* = \{f_0, f_2\}$ and is the only feature with exact minimum marginal $-1$. Algorithm A samples this witness with probability $p_- = 1/[5\binom{4}{2}] = 1/30$. For illustration only, condition Algorithm B's perturbation kernel on the oracle seed $S^*$ rather than on learned MI-correlation guidance. Its probability of returning exactly $S^*$ is $\rho = .49/5 + 2[.21/(5 \cdot 3)] + .09/(5 \cdot 6) = .129$. Figure 2(b) therefore isolates the perturbation kernel; it is not real-data evidence that learned guidance is better.

### 2.4 Quota-Indexed Selection and Validation Gate

For each quota, the estimated harmful set $\widehat{\mathcal{H}}_\alpha = \{i : \widehat{\mathrm{MinPPI}}_i = -1\}$ proposes one compression–utility operating point. Sweeping $\alpha \in \{.85, .90, .95, .99\}$ does not solve a Pareto optimization problem and need not produce monotone or nondominated points. The formal removal rule uses the prespecified default $\alpha = .95$.

Every coalition model is fitted on the training split and its game utility is evaluated on validation data. The same validation split is therefore reused adaptively to construct the harmful set, apply the removal gate, and drive NSGA-II search. This reuse is deliberate but bounded, because every rule applied on validation—the quota, the gate margins, and the all-or-none backoff—is prespecified and frozen, so validation is used to apply fixed decisions rather than to tune free parameters. The held-out split is reserved for registered final evaluation and is never consulted during selection. The candidate subset may trail the full model on validation by at most .005 for classification and .01 for regression. When the raw set fails, the frozen all-or-none backoff removes none of its candidates. The accepted count also defines the deletion budget for SHAP-bottom, MinShap-bottom, Banzhaf-bottom, and random-matched controls. This gate limits validation loss but is not a held-out non-inferiority guarantee.

## 3 Experimental Design

### 3.1 Data, Models, and Scope

Table 1 summarizes the six benchmark datasets. Each seed uses a 64/16/20 train/validation/test split, stratified for classification. Variance filtering, mutual-information selection to at most 100 features, and standardization are fitted on training data only. The frozen seeds are $42, 123, 456, 789, 1024$.

Table 1: Datasets and configured dimension cap. The realized analyzed dimension can be smaller after training-only variance filtering.

| Dataset | Native $p$ | Cap | Samples | Task / utility |
|---|---|---|---|---|
| Spambase | 57 | 57 | 4,601 | binary / ROC AUC |
| Madelon | 500 | 100 | 2,600 | binary / ROC AUC |
| ISOLET | 617 | 100 | 7,797 | 26-class / OVR AUC |
| Gisette | 5,000 | 100 | 7,000 | binary / ROC AUC |
| Credit-Card-Fraud | 29 | 29 | 284,807 | binary / ROC AUC |
| Abalone | 8 | 8 | 4,177 | regression / $R^2$ |

All tasks use XGBoost with probabilities for classification, 100 boosting rounds, depth 4, learning rate .1, one CPU worker per fit, and the CUDA tree backend. The 100-round choice was selected from $\{10, 20, 40, 100\}$ using training/validation-only calibration before formal test evaluation. Binary datasets use ROC AUC, ISOLET uses one-vs-rest AUC, and Abalone uses $R^2$; ISOLET accuracy appears only in a seed-42 utility ablation.

Removal and full-versus-prefiltered NSGA-II cover all six benchmark datasets and five seeds. Both NSGA-II (Deb et al., 2002; Hamdani et al., 2007) variants minimize $|S|/p_0$ and $1 - \mathrm{clip}(U, 0, 1)$, where $p_0$ is the realized analyzed dimension before optional prefiltering and is shared by both variants. They use population 40, 800 evaluations, and reference point $(1.1, 1.1)$. Estimator budgets are $\{2, 5, 10, 25\}$, except ISOLET, which omits 25; exact enumeration is restricted to at most 16 features and therefore provides formal ground truth only on Abalone. The registered $\alpha$ sweep covers all six datasets at seed 42. Guidance and dispatcher ablations cover Madelon, ISOLET, Credit-Card-Fraud, and Abalone at seed 42; the utility ablation is restricted to ISOLET at seed 42.

### 3.2 Comparisons, Statistics, and Provenance

The removal comparison includes full, MinPPI, SHAP-bottom (Lundberg & Lee, 2017), MinShap-bottom (Zheng & Raskutti, 2026), Banzhaf-bottom (Karavas et al., 2017), and random-matched subsets. Thirty random-matched subset trials are averaged within seed, preserving five paired seed-level replicates; each trial draws $k$ features uniformly without replacement, where $k$ is the accepted MinPPI-matched removal count, deterministically seeded by the dataset–seed unit and trial index, and the recorded subsets are re-derived and fail closed on any deviation. Dataset-level tables report means and sample standard deviations. Across datasets, removal ranks use each method's five-seed dataset mean; a Friedman test and paired Wilcoxon tests with Holm adjustment are descriptive because there are only six blocks.

NSGA-II endpoints are validation hypervolume, held-out hypervolume of the validation-selected front, and a post-hoc attainment count: the first evaluation at which cumulative-best validation hypervolume reaches 95% of that run's final-best value. This diagnostic is neither a stopping time nor attainment of a common quality target. Exact paired sign-flip tests align the five seed replicates and rely on descriptive seed-level sign exchangeability; Holm correction is applied within each seven-comparison endpoint family (six datasets plus the dataset-averaged result). The campaign completed every registered job in the five-seed core stages and the seed-42 sensitivity stages, with no approved omissions or skipped jobs. Protocol and provenance details are given in the Reproducibility Details in the appendix.

## 4 Results

### 4.1 RQ1: Validation-Gated Removal

Table 2 shows heterogeneous compression. Averaged over all 30 dataset–seed units, MinPPI removed 12.53% of analyzed features; the descriptive mean held-out $\Delta U$ was $-.000313$ across heterogeneous AUC and $R^2$ scales, and 14 of 30 units removed at least one feature. Madelon had the largest mean compression (43.6%) and a positive mean $\Delta U = +.00469$, whereas Abalone compressed by 27.5% with mean $\Delta U = -.00565$; Credit-Card-Fraud compressed by 2.8% with held-out $\Delta U = -0.001$. The raw harmful set passed its validation gate in 27 of 30 units, including

13 vacuous passes with no proposed removal; among the 17 nonempty candidate sets, 14 passed. The failed cases backed off to zero removal, demonstrating that the gate is a substantive part of the procedure.

Table 2: MinPPI removal at $\alpha = .95$ over five seeds. Compression and held-out $\Delta U = U_{\text{selected}} - U_{\text{full}}$ are mean $\pm$ sample SD. "Raw pass" counts validation-gate passes before backoff, including vacuous passes when no removal was proposed.

| Dataset | Compression (%) | Held-out $\Delta U$ | Raw pass |
|---|---|---|---|
| Abalone | $27.50 \pm 25.62$ | $-.00565 \pm .01077$ | $3/5$ |
| Credit-Card-Fraud | $2.76 \pm 2.89$ | $-.00078 \pm .00335$ | $5/5$ |
| Gisette | $.60 \pm .89$ | $-.00011 \pm .00016$ | $5/5$ |
| ISOLET | $.00 \pm .00$ | $.00000 \pm .00000$ | $5/5$ |
| Madelon | $43.60 \pm 31.96$ | $+.00469 \pm .00497$ | $4/5$ |
| Spambase | $.72 \pm .99$ | $-.00003 \pm .00007$ | $5/5$ |

Table 3: Five-seed mean held-out utility achieved by each removal variant after deleting the accepted MinPPI-matched count of features. Utility is ROC AUC, one-vs-rest AUC, or $R^2$ by task; higher is better, and values are comparable only within a row. "Full" is the full-model utility, and all variants on a dataset use the same deletion budget. SHAP, MinShap, Banzhaf, and Random abbreviate SHAP-bottom, MinShap-bottom, Banzhaf-bottom, and random-matched; Credit abbreviates Credit-Card-Fraud.

| Dataset | Full | MinPPI | SHAP | MinShap | Banzhaf | Random |
|---|---|---|---|---|---|---|
| Madelon | .7876 | .7923 | .7985 | .7861 | .7884 | .7144 |
| Abalone | .5397 | .5340 | .5333 | .5251 | .5174 | .5027 |
| Credit | .9734 | .9726 | .9728 | .9736 | .9736 | .9736 |
| Gisette | .9887 | .9886 | .9887 | .9889 | .9887 | .9887 |
| ISOLET | .9935 | .9935 | .9935 | .9935 | .9935 | .9935 |
| Spambase | .9858 | .9858 | .9858 | .9856 | .9859 | .9858 |

Table 3 lists the held-out utilities achieved by the removal variants as a baseline comparison on the same deletion budget.

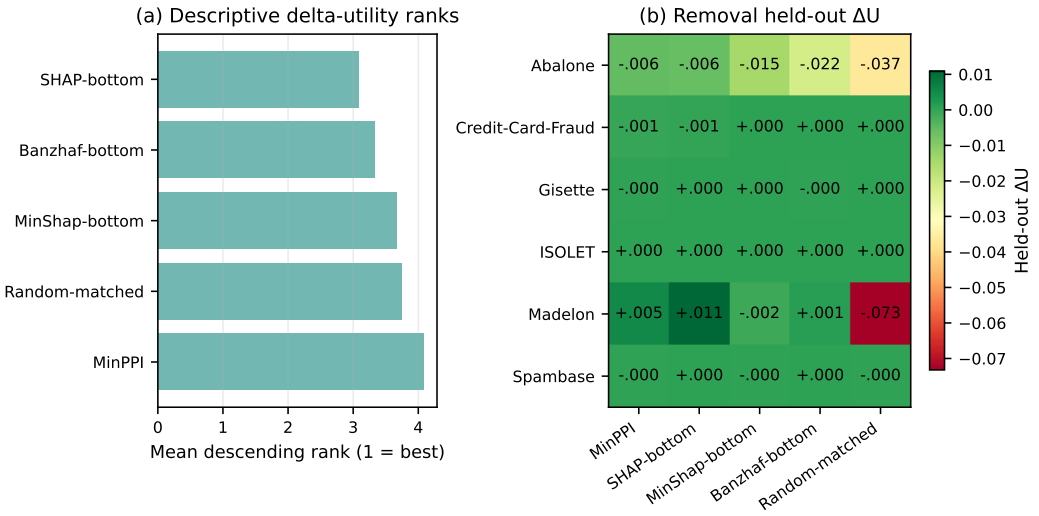



Figure 1: Validation-gated matched removal. (a) Descriptive mean descending ranks (1 is best) among all six methods across six benchmark datasets, computed from each method's five-seed mean $\Delta U$; the full model participates in the ranking but is omitted from the display. (b) Five-seed mean $\Delta U$, where positive values favor removal. Each control uses the accepted MinPPI-matched removal count, and 30 random-matched subset trials are averaged within seed. The display is descriptive and does not establish a broad method ordering.

The descriptive, unweighted mean of the six dataset-level utilities was .877807 for MinPPI and .878121 for the full model, a difference of $-.000313$ (Wilcoxon $p = .4375$, Holm $p = 1$); because ROC AUC, OVR AUC, and $R^2$ are mixed, this mean has no single common utility unit. MinPPI's mean rank among all six variants was 4.08. The omnibus Friedman test found no evidence of a method effect ($\chi^2 = 1.686$, $p = .891$), and every MinPPI pairwise Holm-adjusted $p$-value was 1. Figure 1 therefore visualizes effect heterogeneity rather than a critical-difference or superiority result. Random matched deletion was particularly unfavorable on Madelon ($\Delta U = -.0732$), but the formal ranking does not establish MinPPI as globally best.

Together these results characterize MinPPI as a conservative feature-deletion strategy whose value scales with the amount of removable structure in the data. On Madelon, whose 500 features are dominated by irrelevant variables, MinPPI removed $43.6\%$ of analyzed features with a positive held-out $\Delta U$, indicating that it identifies genuinely harmful features. On ISOLET, where all 617 features carry signal, it correctly found nothing to remove. On Credit-Card-Fraud, Gisette, and Spambase—datasets with mostly relevant features—removals were small and near-lossless. The cautionary case is Abalone, a small regression dataset with a noisy target and only eight features, where MinPPI's $27.5\%$ removal slightly reduced held-out utility, suggesting that weakly informative features can be misclassified as harmful. The validation gate is a substantive safeguard in exactly this setting, vetoing the raw harmful set in the two Abalone seeds and one Madelon seed whose proposed removals failed the validation-loss bound. Compression is also seed-sensitive (standard deviations of roughly 25–32 percentage points on Madelon and Abalone), so the harmful set should be read as a five-seed aggregate rather than a single-run decision.

### 4.2 RQ2: Quota Sensitivity

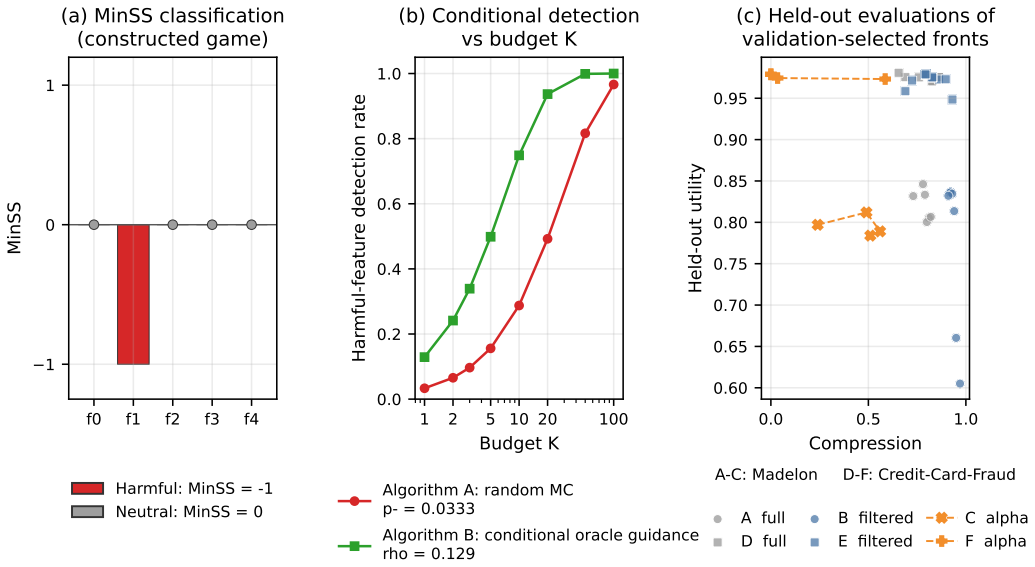



Figure 2: Overview of MinPPI. (a) Exact minimum marginals in the five-feature game of equation 6: $f_1 = -1$ and all other features are neutral at the worst-case level. (b) Geometric detection probabilities for the unique harmful witness under random MC ($p_- = 1/30 = .0333$) and Algorithm B's perturbation kernel conditional on the oracle seed $\{f_0, f_2\}$ ($\rho = .129$); this isolates the kernel and is not real-data guidance evidence. (c) Held-out evaluations of seed-42 validation-selected fronts for full and gated prefiltered NSGA-II, together with the registered $\alpha \in \{.85, .90, .95, .99\}$ operating points on Madelon and Credit-Card-Fraud. Connecting lines only order the $\alpha$ values and do not assert held-out nondominance.

The $\alpha$ sweep is a seed-42 sensitivity analysis ($n = 1$ per point), not a model-selection exercise. Its trajectories were dataset-dependent and non-monotone. On Madelon, compression across $\alpha = (.85, .90, .95, .99)$ was $(.24, .49, .56, .51)$ with held-out $\Delta U = (+.0077, +.0227, -.0003, -.0054)$. On Abalone, compression was $(.375, .50, .50, 0)$, while on Credit-Card-Fraud it rose from 0 at .85 and .90 to .034 at .95 and .586 at .99. Figure 2(c) overlays the registered points for Madelon and Credit-Card-Fraud on held-out evaluations of their seed-42 validation-selected fronts; the complete six-dataset sweep is reported in Table 4 in the appendix. The points illustrate quota sensitivity only; they do not prove monotonicity, held-out nondominance, or cross-seed safety.

### 4.3 RQ3: Prefiltered Multi-Objective Search

With the same 800-evaluation budget, MinPPI-prefiltered NSGA-II changed the dataset-averaged held-out hypervolume by $+.01584$ and validation hypervolume by $+.01741$. For each endpoint, all five seed-level pooled differences were positive (rank-biserial correlation 1), but the exact sign-flip $p = .0625$ became .4375 after the registered Holm correction. Dataset mean test-HV differences were $+.09515$ on Madelon, $+.00535$ on Credit-Card-Fraud, $+.00021$ on Abalone, 0 on ISOLET, $-.00191$ on Gisette, and $-.00378$ on Spambase. Thus the positive pooled direction is driven mainly by Madelon, with only small positive differences on Credit-Card-Fraud and Abalone.

The post-hoc 95%-of-final-HV attainment diagnostic occurred 22.67 evaluations earlier on average ($p = .4375$, Holm $p = 1$), with the largest decreases on Madelon and Gisette and an increase on Spambase. Both variants nevertheless consumed the same registered total budget of 800 evaluations. Figure 3 therefore supports a directional, dataset-dependent change in search outcomes, not universal improvement, budget reduction, or runtime acceleration.

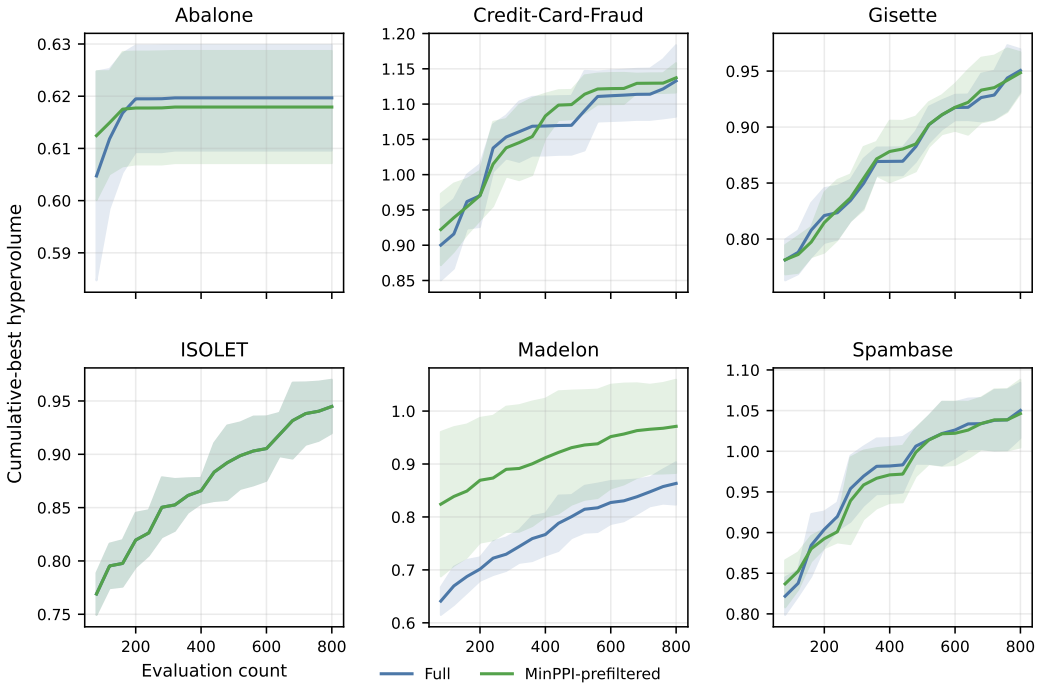



Figure 3: Hypervolume convergence of full versus validation-gated MinPPI-prefiltered NSGA-II under the matched 800-evaluation budget. Each panel is a dataset; blue and green lines are the five-seed means of cumulative-best validation hypervolume for the full and MinPPI-prefiltered variants, with shading $\pm$ one seed-level standard deviation and reference point $(1.1, 1.1)$. Prefiltering leaves the curves essentially coincident on most datasets and produces a sustained improvement on Madelon; the dataset-averaged held-out hypervolume change is +0.016. Both variants consume the same 800 evaluations, and no dataset-wise comparison survives the registered exact paired sign-flip tests with within-endpoint Holm correction.

### 4.4 RQ4: Estimator, Guidance, and Utility Sensitivity

Abalone is the only dataset small enough for exact enumeration. Averaged over the five seed-level F1 scores, at budgets $2, 5, 10, 25$, random MC achieved $.100, .150, .230, .443$, while MI-correlation guidance achieved $.000, .080, .133, .313$. The hybrid dispatcher scored 1 at every budget because it selected exact enumeration at eight features; this does not validate the approximate guidance heuristic. Thus the five-seed mean F1 for learned guidance was lower than random sampling at every registered finite budget; this is not a claim about every individual seed. Table 5 in the appendix reports the full estimator comparison.

The seed-42 ablations likewise show sensitivity rather than a stable guidance ordering. On Madelon, random, MI-only, correlation-only, and MI-correlation seeds recovered 5, 16, 34, and 28 harmful labels; on Credit-Card-Fraud the counts were 6, 2, 2, and 1. On ISOLET, changing the utility while keeping the probability-producing classifier fixed changed the harmful count from 0 under the formal OVR AUC to 9 under accuracy. These single-seed results show that estimator and utility choices materially affect recovered labels, but they cannot support significance or cross-seed stability claims. Table 6 summarizes the seed-42 harmful counts.

A practical reading follows. Exact enumeration and the hybrid dispatcher recover every harmful label on Abalone, so MinPPI's reliable value lies in the exact or hybrid path when the dimension is small enough; the MI-correlation guidance is best treated as a fallback for larger problems, where it is not validated to improve on random sampling. The sensitivity of the recovered harmful set to estimator, seed construction, and utility likewise reinforces that MinPPI should be consumed as a conservative, gate-protected, multi-seed aggregate rather than a single configuration.

## 5 DISCUSSION

The exact contribution of the MinPPI specialization is sharper than its empirical claim. Given $\alpha$, an exact or sampled $-1$ is existence-certified by an evaluated threshold-crossing coalition and needs no additional score threshold; the validation gate then separates witness discovery from the decision to remove a feature. Finite sampling preserves witness validity but not witness completeness, and sampled 0 or $+1$ does not certify the corresponding exact class. MinPPI is a worst-case support statistic rather than an interaction index: unlike Shapley interaction indices (Grabisch & Roubens, 1999) and their Shapley–Owen specialization (Rabitti & Borgonovo, 2019), which quantify feature interactions beyond single-feature attribution, MinPPI certifies a single harmful threshold-crossing witness per feature. This distinction is important because this study does not support a universal selector ranking: MinPPI stayed close to full-model utility descriptively, yet had mean rank 4.08 among six methods, and Abalone exhibited a negative mean held-out effect.

The downstream results are similarly conditional. Madelon accounts for most of the positive average hypervolume change; Credit-Card-Fraud and Abalone show only small positive differences, while Gisette and Spambase show small decreases. This pattern is compatible with prefiltering being useful when its accepted set removes distracting dimensions, but the study neither identifies a general dataset condition nor establishes statistical superiority. The registered validation gate also cannot guarantee held-out non-inferiority; it uses no test information by design.

These findings delimit where MinPPI is most useful. It is best read as a conservative feature-deletion strategy: it removes only features certified as harmful by a threshold-crossing witness, subject to a validation gate that vetoes any removal failing a fixed loss bound. This conservatism concentrates its strongest results on datasets with abundant removable structure—Madelon, where it removed $43.6\%$ of features at negligible cost and improved downstream search—while it correctly removes little or nothing where all features carry signal, and it needs the gate most on small noisy-target problems such as Abalone. The natural applications are (i) prefiltering high-dimensional feature spaces before modeling or multi-objective search, (ii) auditing feature quality by flagging features with harmful witnesses, which can expose suspicious, noisy, or leak-prone predictors, and (iii) a witness-certified, gate-protected removal step for single-task models that avoids blind deletion in multi-task settings. MinPPI therefore does not require perfectly clean data; it requires data in which removable structure exists, and its output should be consumed as a five-seed aggregate subject to the validation gate.

Several limitations constrain interpretation. First, five paired seed replicates and six benchmark datasets give low-powered tests; seed-level sign-flip inference is descriptive and depends on sign exchangeability. The $\alpha$, guidance, dispatcher, and utility ablations use only seed 42. Second, the study uses one XGBoost family and utilities on different scales; a common multiplicative quota is not invariant to shifts of the utility scale. Third, exact estimator comparison is available only on eight-feature Abalone, where the five-seed mean F1 for guidance was lower than random sampling at every registered finite budget. The bound in equation 5 is a support condition, not evidence that the guidance distribution assigns more mass to harmful witnesses. Fourth, shared caching and execution order complicate runtime attribution, so we make no computational speedup claim. Finally, the current figures summarize means and sample standard deviations; paired seed-level variation, broader model families, and a stronger learned-guidance estimator for finite-budget harmful-feature detection remain important future work.

## 6 CONCLUSION

Exact MinPPI specializes minimum marginal aggregation to a binary utility-threshold game and yields a quota-conditioned three-way classification; finite-budget estimators certify only observed harmful threshold crossings. The resulting candidates feed a validation-gated removal or prefiltering step. Across the six benchmark datasets, the method produced nonzero and sometimes substantial compression on some datasets and a positive descriptive average NSGA-II hypervolume change, but neither removal superiority nor a general guidance advantage was detected. The appropriate conclusion is therefore methodological and conditional: MinPPI is a conservative feature-deletion strategy whose strongest performance is concentrated on datasets with abundant removable structure, and whose value otherwise depends on the model, split, dataset, estimator, utility, quota, and validation

decision—consistent with the observation that harmful features are context-dependent rather than universally removable (Lippl & Lindsey, 2024; Siam et al., 2026).

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# A PROOFS

*Proof of Proposition 1.* Because $v_\alpha \in \{0,1\}$, every marginal $\Delta_i(S)$ is in $\{-1,0,+1\}$, and so is any nonempty minimum of such marginals. For each $S \subseteq N \setminus \{i\}$ there are $|S|!(n-|S|-1)! \geq 1$ permutations for which $S$ is exactly the predecessor set of $i$. Hence the distinct arguments of $\mathrm{MinSS}_i$ and $\mathrm{MinBz}_i$ coincide; multiplicity cannot change a minimum.

The minimum equals $-1$ if and only if at least one marginal is $-1$, which is equivalent to the existence of $S$ with $v_\alpha(S) = 1$ and $v_\alpha(S \cup \{i\}) = 0$. It equals $+1$ if and only if every marginal is $+1$, which is equivalent to $v_\alpha(S) = 0$ and $v_\alpha(S \cup \{i\}) = 1$ for every $S$. If neither condition holds, the trinary range forces the minimum to be 0. □

*Finite-sample one-sidedness.* For any sampled collection $\mathcal{S}_K \subseteq 2^{N\setminus\{i\}}$,

$$\min_{S\in\mathcal{S}_K} \Delta_i(S) \geq \min_{S\subseteq N\setminus\{i\}} \Delta_i(S),$$

because the left minimum is taken over a subset of the right minimum's arguments. An observed value $-1$ is therefore exact evidence of at least one harmful witness, but a larger sampled minimum can result from an unseen smaller marginal. □

*Conditional detection bound.* Let independent perturbations be drawn from $P_i$, and suppose a fixed harmful witness $S_i^*$ has mass $\rho_i > 0$. One draw misses that witness with probability $1-\rho_i$; all $K_{\mathrm{adv}}$ draws miss it with probability $(1-\rho_i)^{K_{\mathrm{adv}}}$. Sampling $S_i^*$ implies detection, which yields equation 5. The statement is silent when $\rho_i = 0$ and does not compare $P_i$ with the random-MC distribution. □

# B ADDITIONAL FORMAL RESULTS

## B.1 QUOTA SWEEP

Table 4 reports the seed-42 quota sweep for the six benchmark datasets. Compression is the accepted removed fraction and $\Delta U$ is held-out selected-minus-full utility. The non-monotone rows show why $\alpha = .95$ was fixed before the campaign rather than chosen retrospectively.

Table 4: Seed-42 quota operating points, shown as (compression, held-out $\Delta U$). Every cell has $n = 1$.

| Dataset | $\alpha = .85$ | $\alpha = .90$ | $\alpha = .95$ | $\alpha = .99$ |
|---|---|---|---|---|
| Abalone | $(.375, +.0041)$ | $(.500, -.0198)$ | $(.500, +.0010)$ | $(0, 0)$ |
| Credit-Card-Fraud | $(0, 0)$ | $(0, 0)$ | $(.034, -.0045)$ | $(.586, -.0057)$ |
| Gisette | $(0, 0)$ | $(0, 0)$ | $(0, 0)$ | $(.090, +.0002)$ |
| ISOLET | $(0, 0)$ | $(0, 0)$ | $(0, 0)$ | $(.010, -.000005)$ |
| Madelon | $(.240, +.0077)$ | $(.490, +.0227)$ | $(.560, -.0003)$ | $(.510, -.0054)$ |
| Spambase | $(0, 0)$ | $(0, 0)$ | $(0, 0)$ | $(.196, -.0012)$ |

### B.2 Estimator and Seed-42 Ablations

Table 5: Harmful-label F1 against exact Abalone labels over five seeds. Exact and hybrid equal 1 at all budgets because the hybrid dispatches to exact enumeration at eight features.

| Estimator | $K = 2$ | $K = 5$ | $K = 10$ | $K = 25$ |
|---|---|---|---|---|
| Random MC | .100 | .150 | .230 | .443 |
| MI-correlation guided | .000 | .080 | .133 | .313 |
| Exact | 1.000 | 1.000 | 1.000 | 1.000 |
| Hybrid dispatcher | 1.000 | 1.000 | 1.000 | 1.000 |

Table 6: Seed-42 ablation harmful counts. A dash means exact enumeration was disallowed by the registered dimension limit, not that zero harmful features were found.

| Axis | Variant | Abalone | Credit | ISOLET | Madelon |
|---|---|---|---|---|---|
| Guidance | random | 2 | 6 | 1 | 5 |
| | MI only | 2 | 2 | 0 | 16 |
| | correlation only | 2 | 2 | 0 | 34 |
| | MI-correlation | 2 | 1 | 0 | 28 |
| Dispatcher | exact | 4 | – | – | – |
| | random MC | 2 | 2 | 0 | 34 |
| | MI-correlation | 2 | 1 | 0 | 28 |
| | hybrid | 4 | 1 | 0 | 56 |

For ISOLET, the registered utility ablation found 0 harmful features with formal OVR AUC and 9 with accuracy. The classifier family and probability outputs were held fixed; the difference is therefore attributable to the utility definition within this seed-42 comparison.

## C Reproducibility Details

The formal splits contain 64% training, 16% validation, and 20% test observations. Dataset identities, split indices, transformation metadata, model policy, source identity, and the scientific software environment are recorded with cryptographic digests. Resume operations reject protocol drift. Training data fit preprocessing, guidance, and every coalition model. Validation utility is reused adaptively for threshold-game construction, the removal gate, and NSGA-II selection. The held-out partition is not used to choose features, the removal backoff, $\alpha$, or NSGA-II individuals; it is read only for registered outcome evaluation after validation selection. Tables and figures are rederived from canonical row-level records and independently hash-verified.

The formal estimator set comprises exact enumeration where permitted, random MC, MI-correlation guidance, and the hybrid dispatcher. Removal controls delete the accepted MinPPI-matched count. NSGA-II-full and NSGA-II-prefilter use identical population, variation, evaluation budget, reference point, and validation-selection logic. Runtime, cache-hit, and physical-fit counters are diagnostic only and are excluded from predictive superiority claims.